

Chapter 1- Fundamentals

Set - set of elements, Core maths structure.

- finite set size is denoted as $|X|$, X is a finite set
- Set builder notation $S = \{ \text{expression} \mid \text{Rule} \}$
- Special sets are denoted with special characters

Cartesian product - product of 2 sets A, B , output is a new set C which contains ordered pairs $C = \{ (x, y) \mid x \in A, y \in B \}$

Note | Product of 2 finite sets A, B has size $|A| \times |B|$

Cartesian power - for any set A , A^n is equal $\{ (x_0, x_1, \dots, x_n) \mid (x_0, x_1, \dots, x_n) \in A \}$

Example | $\mathbb{Q}^3 = \{ (q_1, q_2, q_3) \mid (q_1, q_2, q_3) \in \mathbb{Q} \} \rightarrow \text{Size} = |\mathbb{Q}|^3 \mid \left(\frac{|\mathbb{Q}|^3}{3} \right) \text{Triplets}$

Subsets - A is subset of B if A has it's elements inside B , $A \subset B$

Fact - $\emptyset$ is always a subset of any set

Fact - Any set has the Number of subsets $= 2^{|S|}$, S is the set

Example | $S = \{a, b\}$, How many subsets does S have these are called "power sets"

$$S = \{a\}, \{\emptyset\}, \{b\}, \{a, b\}$$

Example | unit circle $C = \{ (x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1 \}$

Graph of function $= \{ (x, f(x)) \in \mathbb{R}^2 \mid x \in \mathbb{R} \}$

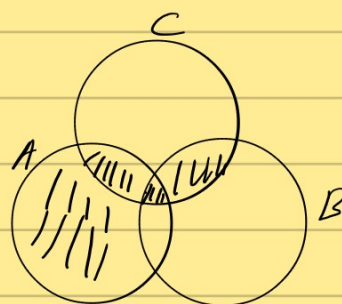
power sets - subset of a set, denoted by $\mathcal{P}(A)$, A is the main set

Sets logic

- Union - $A \cup B \rightarrow \{ x \mid x \in B \text{ or } x \in A \}$
- Intersection - $A \cap B \rightarrow \{ x \mid x \in B \text{ and } x \in A \}$
- Difference - $A - B \rightarrow \{ x \mid x \in A \text{ and } x \notin B \}$
- Complement - $A' / \bar{A} \rightarrow \{ x \mid x \notin A \text{ and } x \in U \}$
- Inclusive OR - $A \cup B$, Inclusive AND - $A \cap B$

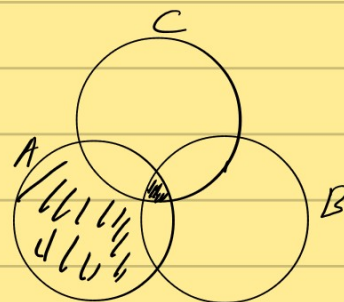
Problem | Describe the following Venn diagram (No repetition)

$$(A \cap B^c) \cup (A \cap B \cap C) \cup (A^c \cap C \cap B)$$



Problem #2 | Describe the following Venn diagram

$$(A \cap C \cap B^c) \cup (A \cap B \cap C)$$



Note | Set logic is just logical expressions for element E

- $A \cap B$ (if E in A And E in B) $\rightarrow E$
- $A \cup B$ (if E in A OR E in B) $\rightarrow E$
- $A \cap B^c$ (if E in A And Not in B) $\rightarrow E$
- $A \cup B^c$ (if E in A Or Not in B) $\rightarrow E$ + Any element Not in $B \in U$

Indexed sets — Sets that have an index $A_1, A_2, A_3, \dots$

Indexed sets operations — $A_1 \cup A_2 \cup A_3 \rightarrow \{x \in A_n \mid x \text{ is an element of at least } A_n\}$

$A_1 \cap A_2 \cap A_3 \rightarrow \{x \in A_n \mid x \text{ is an element of every } A_n\}$

$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \dots A_n$ shortcut for indexed sets operations

$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \dots A_n$

More Indexed sets operations:

$\bigcup_{i \in \mathbb{N}} A_i \rightarrow A_1 \cup A_2 \cup \dots$ there is at least one element that satisfy $i \in \mathbb{N}$

$\bigcap_{i \in \mathbb{N}} A_i \rightarrow A_1 \cap A_2 \cap \dots$ there is $i \in \mathbb{N}$ in every set?

Well ordering principle — if a set is a subset of a special set, then element x_0 must be the smallest element.

Division Algorithm — Given integers A, B , ($B \in \mathbb{N}$). the division of A by B is defined by

$A = qB + r$, A is the dividend, B is the divisor, q is the quotient, r is the remainder

Note | $x/y = q \mid x - qy = r$

Logic - Its the process of combining existing information to produce New information, theorms are proven with logic

Statements - They are mathematical sentences / information, that can be either true or false they can be denoted by letters, P , they can contain variables $P(X)$. Any true result or theorm is also a statement.

Open sentences - Statements that are true or false based on a given variable $P(X)$: X is an even number.

Combining Statements - S_1, S_2 (truth values)

- And, $S_1 \wedge S_2 \rightarrow$ Both must be true to give S_3
- Or, $S_1 \vee S_2 \rightarrow$ Either can be true to give S_3
- Negation / Not, $\neg S_1 \rightarrow$ if S_1 is T then it becomes F and vice versa $\leftarrow$ requires only 1 statement
- If condition, $S_1 \Rightarrow S_2$, if S_1 then S_2 " S_1 implies S_2 " S_2 if S_1
- Converse, $S_1 \Rightarrow S_2, S_2 \Rightarrow S_1$, these are 2 converse relations
- Double implication, $S_1 \Leftrightarrow S_2, S_1$ if & only if $S_2 \Rightarrow (S_1 \Rightarrow S_2) \wedge (S_2 \Rightarrow S_1)$
Both can be either true or false for a double implication to occur

Example | $P \Leftrightarrow Q$, P is necessary & a sufficient condition for Q

Example | The product of $X \& Y = 0$ if and only if $X=0 \& Y=0$.

$$(XY=0) \Leftrightarrow (X=0 \wedge Y=0)$$

logical equivilance - Two statements are logically equivalent if their truth values align

De Morgan's law

- $\neg (P \wedge Q) \rightarrow$ We don't need both $P \& Q \rightarrow (\neg P) \vee (\neg Q)$
- $\neg (P \vee Q) \rightarrow$ we don't want either $\rightarrow (\neg P) \wedge (\neg Q)$

Distributive law - just like algebra $a(x+y)$ we distribute $ax + ay$

- $Q \wedge (X \vee Y) \rightarrow (Q \wedge X) \vee (Q \wedge Y)$
- $Q \vee (X \vee Y) \rightarrow (Q \vee X) \vee (Q \vee Y)$

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Associative law - parenthesis are useless when we only have a common logical operand, but it is essential when we have a mix of $\vee$ & $\wedge$

Quantifiers - They sense to quantity

- $\forall$ - for all elements (universal quantifier)
- $\exists$ - Exist an element (existential quantifier)

Example For every $n \in \mathbb{Z}$, $2n$ is even.

$$p(n): (\forall n \in \mathbb{Z}) (2n \text{ is even})$$

Example For every integer N that's not even, it's odd & vice versa

$$p(n): (\forall n \in \mathbb{Z}) (\neg E(n) \Leftrightarrow O(n))$$

Example Every even integer greater than 2 is the sum of 2 primes

$$(\forall E(n) \in \mathbb{Z}, n > 2) (\exists p, p \in \mathbb{P}, p + p = n)$$

Negation of qualifiers:

$$\bullet \neg \forall \rightarrow \exists \text{ "For all, Only one"}$$

$$\bullet \neg \exists \rightarrow \forall \text{ "only one, For all"}$$

Example For all $x \in \mathbb{Z}$, x is even $\rightarrow \forall x \in \mathbb{Z}, E(x)$

There exists an x in $\mathbb{Z}$ that is not even $\rightarrow \exists x \in \mathbb{Z}, \neg E(x)$

Example $\forall x \in \mathbb{R}, x^2 > 0$

$$\exists x \in \mathbb{R}, x^2 \leq 0 \text{ Neg.}$$

Example $(\forall x \in \mathbb{R}, \exists y \in \mathbb{R}) y^3 = x$

$$(\exists x \in \mathbb{R}, \forall y \in \mathbb{R}) (y^3 \neq x) \text{ Neg.}$$

Example If a is odd, a^2 is odd

$$O(a) \Rightarrow O(a^2)$$

$$O(a) \Rightarrow \neg O(a^2) \mid O(a) \wedge \neg O(a^2) \text{ Neg.}$$

Practice For every $\epsilon > 0$, there is a $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$

$$(\forall \epsilon > 0, \exists \delta > 0): (x \in U, a \in U, |x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon)$$

$$(\exists \epsilon > 0, \forall \delta > 0): (x \in U, a \in U, |x - a| < \delta \wedge |f(x) - f(a)| > \epsilon) \text{ Neg.}$$

Practice #2 There exists a real number a for which $a + x = x$ for every real x

$$(\forall x \in \mathbb{R}, \exists a \in \mathbb{R}): (a + x = x)$$

$$(\exists x \in \mathbb{R}, \forall a \in \mathbb{R}): (a + x \neq x)$$

